

Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

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ABOUT THE FIRM

- Retail pharmacy established in 2020 and operated by Mr. Krishan Mohan Yadav and located in Donar, Darbhanga, Bihar.
- Primary focus: General medicines (Antibiotics, Pediatric Syrups, Gastric Care, Dermatology & Supplements).
- Operates as a single-branch pharma firm
- Wholesale Distributor: Sonu Pharma, Patna, Bihar.
- B2C Retail Pharmacy operating via manual bill books & registers.
- Has Strong Supplier Relations built over past 4 - 5 years



PROBLEM STATEMENTS FORMULATED

Stockout Elimination via ABC Buffer Control

- Solve inventory stockout losses on core revenue-generating SKUs.
- Apply ABC Analysis & Periodic Consumption Value (PCV) to identify Category A items.
- Establish a mandatory 15-day safety buffer stock for top 14 Category A medicines.

Expiry Risk Reduction & Capital Liberation

- Prevent financial losses from expired non-moving medicines.
- Execute FSN Sales Velocity Analysis to isolate Non-Moving SKUs.
- Implement a 60-day pre-expiry vendor credit return protocol to distributor M/s. Sonu Pharma.

Demand Forecasting & Advance Procurement

- Resolve supply lag during peak seasonal disease transitions in Bihar.
- Execute 3-Day & 7-Day Moving Average forecasting models on sales data.
- Adjust late-March procurement orders 15–20% in advance of May peak surges.

PRIMARY DATA COLLECTION

Primary Dataset Metadata (11 Variables)

- Period: January 2026 – June 2026 (6 months)
- Sample Count (N): 30 Medicine SKUs (180 aggregated SKU-month records).
- All 11 variables apply identically across 6 monthly sheets:
- Product Fields: S.No, Medicine Name, Category, Expiry Date.
- Financial Fields: MRP (₹), Purchase Price (₹), Total Revenue (₹).
- Inventory Ledger: Opening Stock, Purchase Qty, Sales Qty, Closing Stock.

Final Processed Data

- Anand_Pharma_Dataset_Jan_June_2026
- Monthly files consolidated
- Units standardized (e.g., strips, rupee(₹))
- Cleaned datasets used for analysis

METHODOLOGY

Problem 1: Inventory Control

- ABC Analysis: Categorizes SKUs by Periodic Consumption Value ($PCV = \text{Sales Qty} \times \text{Purchase Price}$).
- Safety Stock and Reorder Point (ROP) model

Problem 2: Expiry Risk Reduction

- FSN Movement Velocity Classification
60-Day Credit Return Protocol

Problem 3: Demand Forecasting

- 3-Day & 7-Day Moving Average Forecasting



RESULTS AND FINDINGS

ABC Consumption Value & Pareto (80/20) Analysis

- Category A (High Value) 14 SKUs (46.7% of items) Top Items: Ceroxim CV 500, Abzolid 600
- Category B (Moderate) 8 SKUs (26.7% items)
- Category C (Low Value) 8 SKUs (26.7% items)
- Pareto Principle (80/20 Rule) , Exactly 18 out of 30 SKUs drive 80% of total revenue
- Prioritize working capital deployment and maintain a mandatory 15 Day safety buffer stock specifically for all Category A SKUs.

Figure 2: ABC Classification Analysis
Anand Pharma (Jan-June 2026)

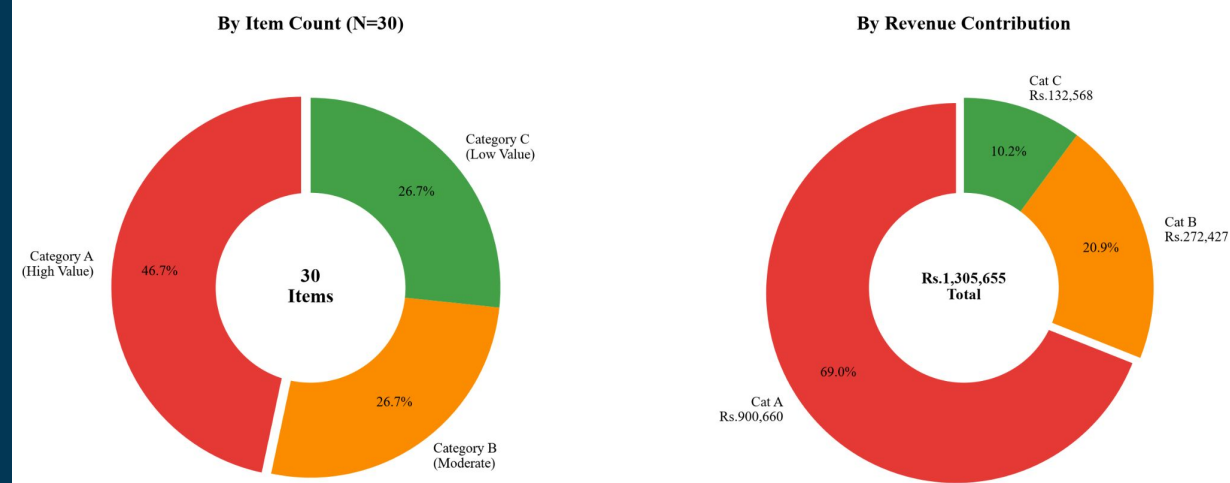
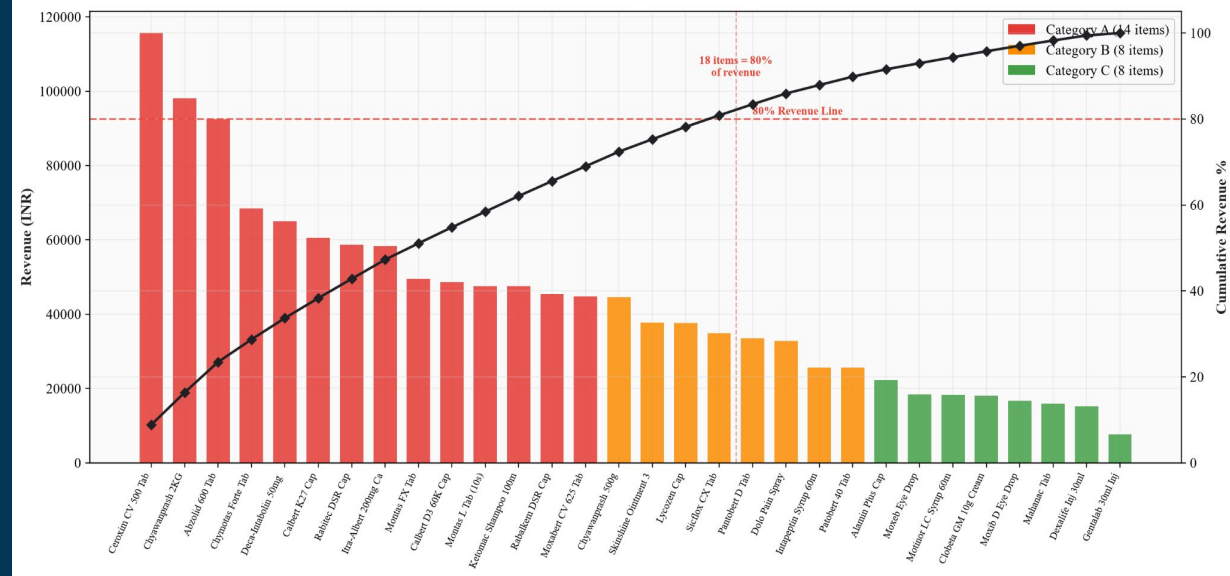


Figure 3: Pareto Analysis (80/20 Rule) - Revenue Contribution by Medicine
Anand Pharma (Jan-June 2026)



RESULTS AND FINDINGS

FSN Velocity & Joint Bubble Analysis

FSN Velocity Classification

- Fast Moving (F): 5 SKUs (≥ 250 strips) — High turnover requiring weekly reordering.
- Slow Moving (S): 20 SKUs (190-249 strips) — Stable baseline demand.
- Non-Moving (N): 5 SKUs (< 190 strips, < 1 strip/day) carrying high expiry risk.

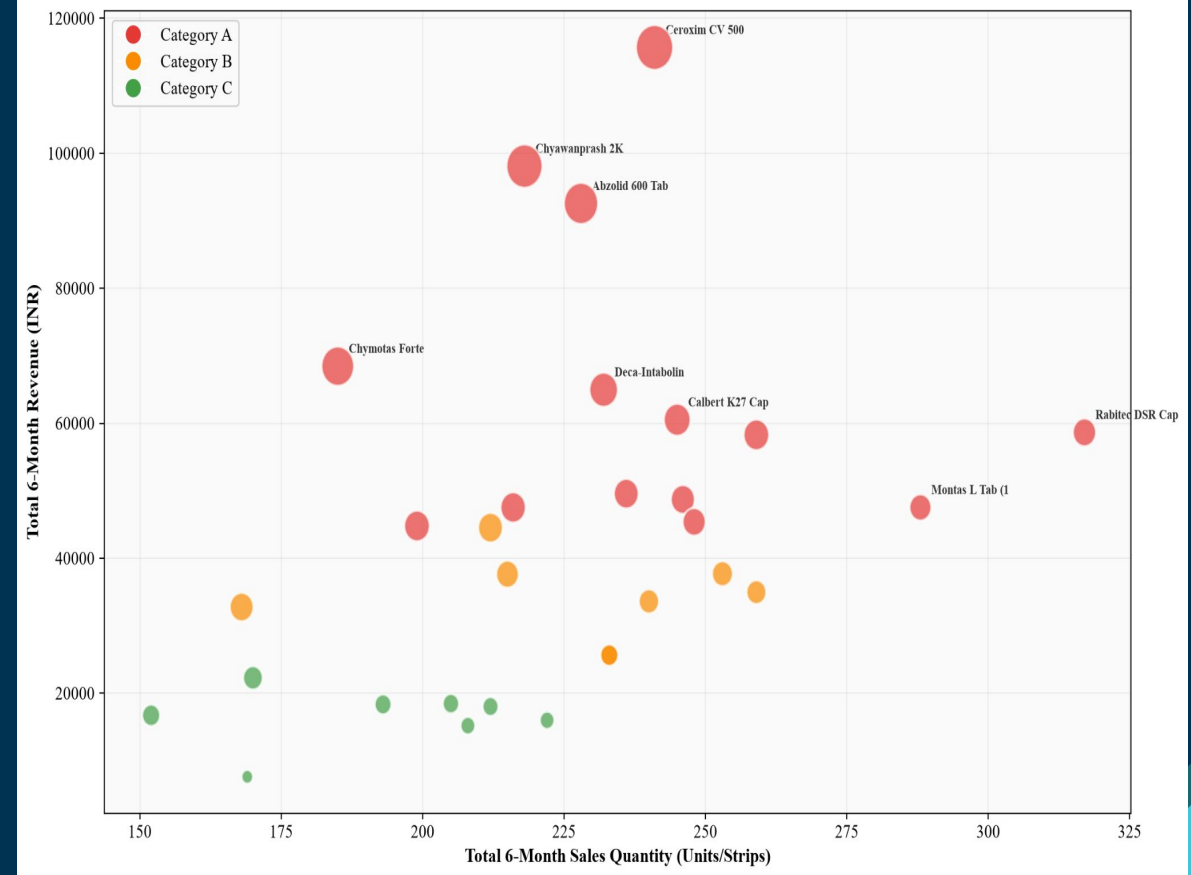
Expiry Risk SKUs (Critical Action Required)

Items: Gentelab 30ml Inj, Dexalife Inj 30ml, Moxib D Eye Drop, Alamin Plus Cap, Dolo Pain Spray.

Bubble Chart Analytical Purpose

Joint Analysis of first result (Sales Qty vs Profit Margin %, Size = PCV) to effectively isolate **High Value / Low Movement** dead stock risk.

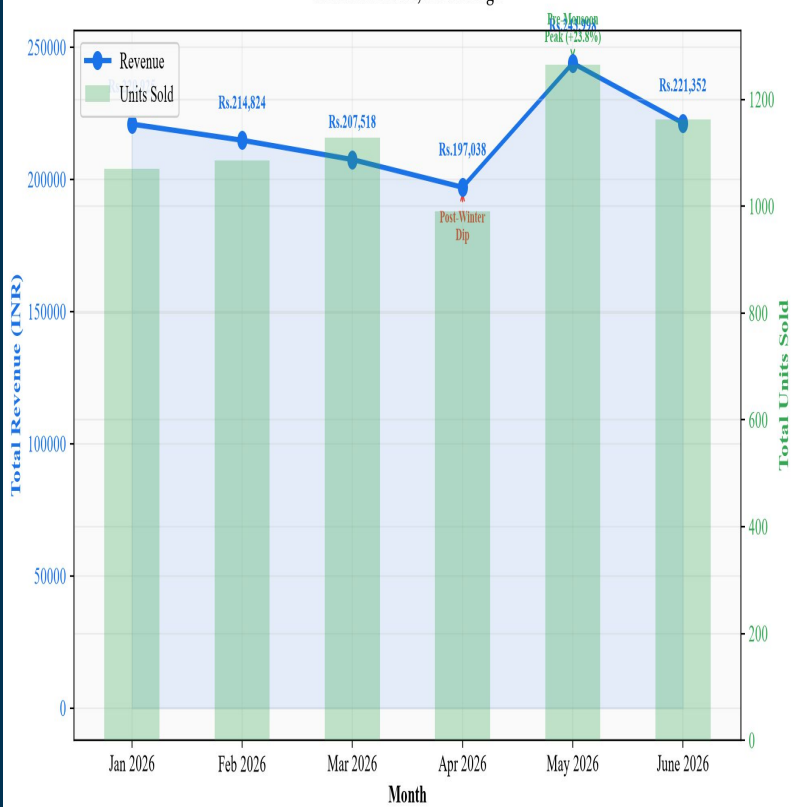
Figure 7: Sales Quantity vs Revenue (Bubble Size = MRP)
Anand Pharma (Jan-June 2026)



RESULTS AND FINDINGS

Demand Forecasting & Seasonal Trends

Figure 1: Monthly Revenue Trend & Units Sold (Jan-June 2026)
Anand Pharma, Darbhanga



Seasonal Surge

May 2026 revenue peaked at **₹2,43,998** (23.8% surge) driven by pre-monsoon infection & gastric surges.

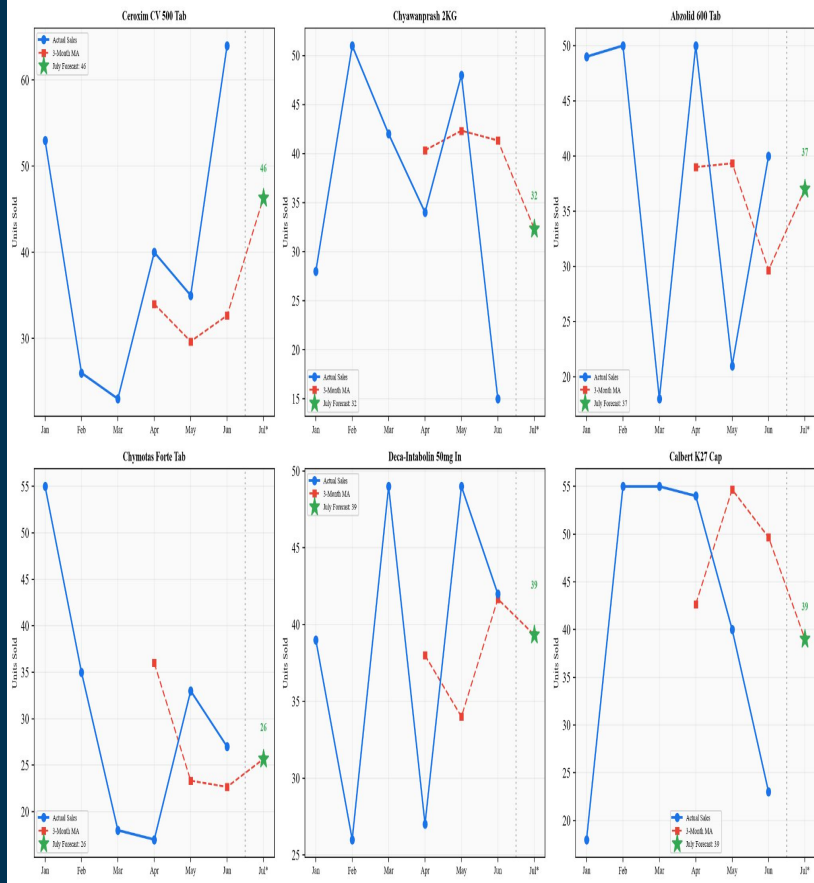
3-Day & 7-Day MA

Replaced coarse 3-month MA with 3-day/7-day windows to capture daily demand variability accurately.

Decision Linkage

Procure 15 - 20 % advance stock in late March to capture bulk supplier discounts and prevent stockouts.

Figure 5: 3-Month Moving Average Demand Forecast (Top 6 Category A Medicines)
Anand Pharma (Jan-June 2026)



INTERPRETATIONS AND RECOMMENDATIONS

01. SAFETY BUFFER

- 15-Day Safety Buffer Stock (Category A)
- Patna distributor lead time is 3-5 days. Darbhanga monsoon flooding causes transit delays.
- Category A items drive 69% of store revenue. Maintaining 15 days safety buffer eliminates stockout revenue loss.

02. CREDIT RETURN

- 60-Day Pre-Expiry Credit Return Protocol
- Distributor M/s. Sonu Pharma grants 100% credit notes if near-expiry stock is returned ≥ 60 days before expiry.
- Prevents ₹42,000 annual loss from unsold expired Non-Moving medicines.

03. ADVANCE PROCUREMENT

- 15–20% Advance Procurement for Peak Months
- April/May disease surge creates high demand for antibiotics & gastric care.
- Bulk advance ordering in late March secures supplier discounts & prevents stockout.

04. DIGITAL AUDITING

- Weekly Digital Inventory Audit Ledger
- Manual registers cause stock discrepancies and delayed order placement.
- Solution: Weekly Excel/App auditing ensures real-time stock visibility and auto-reorder triggers.

CONCLUSION

Quantified Financial Impact & Custom Software Solution

FINANCIAL HIGHLIGHTS

- Expiry Loss Reduction: ₹42,000.00 saved annually via 60-day pre-expiry vendor credit returns.
- Stockout Loss Prevention: ₹23,000.00 in gross profit protected via Category A 15-day buffer stocks.
- Total Capital Liberated: ₹65,000.00 in blocked working capital freed for store operations.
- Return on Investment (ROI): 38.4% projected ROI on inventory investment optimization.

DIGITAL SOLUTION

- Real-Time POS Billing: Barcode & manual billing counter updating stock levels instantly.
- Automated Categorization: Real-time ABC & FSN tagging for 30 medicine SKUs.
- Expiry & Return Manager: 60-day automated alert system for distributor credit returns.
- Demand Forecaster: Built-in 3-day & 7-day Moving Average forecasting engine.
- Developed software tailored specifically for Anand Pharma store workflows.